



**SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS**

**2003
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1**

Mathematics Extension 2

General Instructions

Reading Time – 5 Minutes

Working time – 90 Minutes

Write using black or blue pen. Pencil may be used for diagrams.

Board approved calculators may be used.

Each question is to be returned in a separate bundle.

All necessary working should be shown in every question.

Total Marks – 60

Attempt questions 1 – 3

All questions are of equal value.

Examiner: *A.M.Gainford*

Question 1. (Start a new answer sheet.) (20 marks)

	Marks
(a) For the complex number $z = \sqrt{3} + i$:	
(i) Write down $() \quad z $	2
$() \quad \arg z.$	
(ii) Plot and clearly label each of the following complex numbers on the Argand diagram:	5
$() \quad z$	
$() \quad \bar{z}$	
$() \quad z^2$	
$() \quad iz$	
$() \quad z\bar{z}$	
(b) Sketch the region in the Argand diagram containing all those points z for which: $ \arg z < \frac{\pi}{6}$ and $1 \leq z \leq 4$. (Take special care at boundaries and corners.)	4
(c) In the Argand diagram sketch the locus of z constrained such that $\arg \frac{z - 2i}{z - 1} = \frac{\pi}{4}$.	2
(d) Solve the equation $z^2 + iz + 2 = 0$.	2
(e) (i) Show that $(x - i)$ is a factor of $P(x) = x^4 + x^3 - 11x^2 + x - 12$.	1
(ii) Hence reduce $P(x)$ to linear factors over the complex field.	2
(f) Calculate the complex square roots of $16 - 30i$.	2

Question 2. (Start a new answer sheet.) (20 marks)

Marks

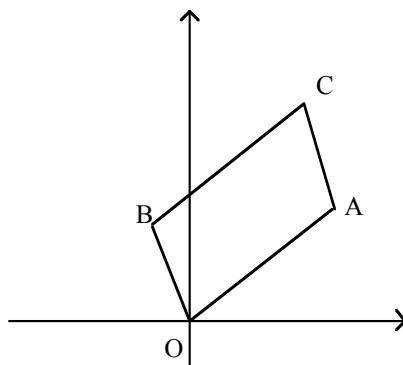
- (a) (i) Show that for any two complex numbers z_1 and z_2 :

1

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

- (ii) In the Argand diagram below A and B represent z_1 and z_2 , and $OACB$ is a parallelogram. Use this diagram to interpret the above result geometrically.

2



- (b) (i) Find the four solutions of the equation $z^4 + 1 = 0$.

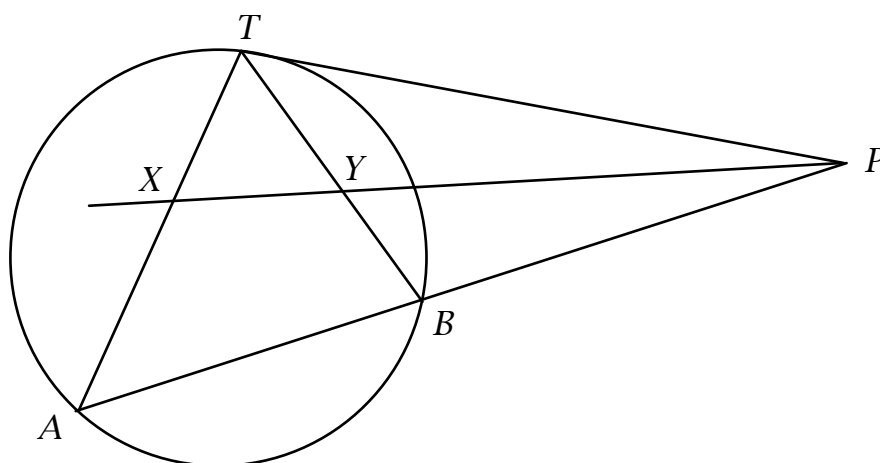
2

- (ii) Hence or otherwise write $z^4 + 1$ as a product of two quadratic factors with real coefficients.

2

- (c)

4



The tangent at T on the circle meets a chord AB produced at P . The bisector of TPA meets TA and TB at X and Y respectively.

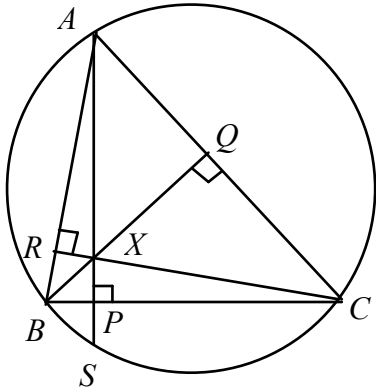
- (i) Give the reason why $\angle PTB = \angle TAB$.

- (ii) Prove $TX = TY$.

- (iii) Prove $\frac{TX}{XA} = \frac{TP}{PA}$.

- (d) (i) Prove that if the polynomial $P(x)$ has a root of multiplicity m , then the derived polynomial $P'(x)$ has a root of multiplicity $m - 1$. 3
- (ii) Find the value of k so that the equation $5x^5 - 3x^3 + k = 0$ has a positive repeated root.
- (e) The origin and the points z , $\frac{1}{z}$, and $z + \frac{1}{z}$ are joined to form a quadrilateral. 3
Write down conditions for z so that the quadrilateral will be
- (i) a rhombus
- (ii) a square
- (f) Containers are coded by different arrangements of coloured dots in a row. The colours used are red, white, and blue. 3
- In an arrangement, at most three of the dots are red, at most two of the dots are white, and at most one is blue.
- (i) Find the number of different codes possible if six dots are used.
- (ii) On some containers only five dots are used. Find the number of different codes possible in this case. Justify your answer.

Question 3. (Start a new answer sheet.) (20 marks)

- | | Marks |
|--|---|
| (a) (i) Use De Moivre's theorem to show that $\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$. | 3 |
| (ii) Hence solve the equation $16x^4 - 16x^2 + 1 = 0$ and deduce the exact values of $\cos \frac{5\pi}{12}$ and $\cos \frac{\pi}{12}$. | 3 |
| (b) If α, β, γ are the roots of the equation $x^3 - 2x + 3 = 0$, find the monic cubic equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$, and hence or otherwise state the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. | 2 |
| <p>(c)</p>  | <p>In the diagram X is the orthocentre of the $\triangle ABC$, which has altitudes AP, BQ, CR. AP produced meets the circumcircle at S.</p> <p>(i) Copy the diagram to your answer sheet.</p> <p>(ii) Prove that $XP = PS$.</p> |
| (d) Shows that $x^3 + ax + b = 0$, where a and b are real numbers, has: | 4 |
| (i) only one real root if $a > 0$ | |
| (ii) two equal roots if $4a^3 + 27b^2 = 0$ | |
| (e) Prove by mathematical induction that | 4 |
| $1 + 2q + 3q^2 + \dots + nq^{n-1} = \frac{1}{(1-q)^2} [(n+1)q^n + nq^{n+1}]$ for all positive integral n . | |

This is the end of the paper.

EXT 2 solutions

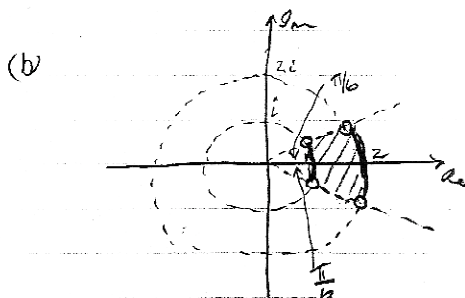
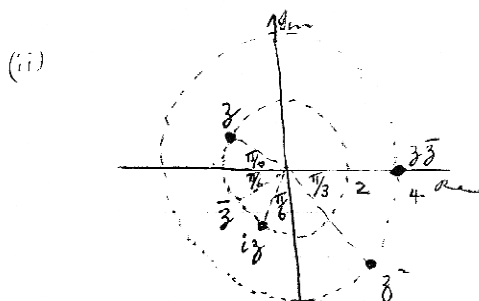
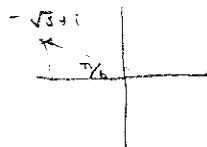
Year 12, 2003, Term 1 Test

Question 1

(a) $z = -\sqrt{3} + i$

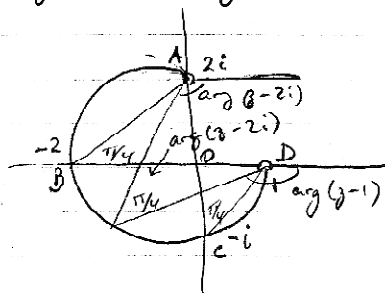
(i) $|z| = \sqrt{(-\sqrt{3})^2 + 1}$
 $= \sqrt{4}$
 $= 2$

(ii) $\arg z = \frac{5\pi}{6}$



(c) $\arg \left(\frac{z-2i}{z-1} \right) = \frac{\pi}{4}$

$\arg(z-2i) - \arg(z-1) = \frac{\pi}{4}$



①

①

⑤

- 1 for rays
- 1 for circles
- 1 for corners
- 1 for boundaries.

④

② 1 for arc of circle
 1 for details

Using isosceles triangles OAB and OCD
 B is -2 C is -i

$\therefore$ center of circle is $(-\frac{1}{2}, \frac{1}{2})$

$r^2 = (\frac{1}{2})^2 + (\frac{3}{2})^2$
 $= \frac{5}{2}$

$\therefore$ The locus is the major arc of the circle $(x+\frac{1}{2})^2 + (y-\frac{1}{2})^2 = \frac{5}{2}$ or thereabouts.

$$\begin{aligned}
 (d) \quad z^2 + iz + 2 &= 0 \\
 \therefore z &= \frac{-i \pm \sqrt{-1 - 4 \times 1 \times 2}}{2} \\
 &= \frac{-i \pm \sqrt{-9}}{2} \\
 &= \frac{-i \pm 3i}{2} \\
 &= i \text{ or } -2i
 \end{aligned}$$

(2)

$$\begin{aligned}
 (e) \quad (i) \quad P(i) &= 1 - i + 11 + i - 12 \\
 &= 0
 \end{aligned}$$

(1)

$\therefore (x-i)$ is a factor of $P(x)$

(ii) As $P(x)$ has real coefficients $x+i$ is also a factor.

$$\begin{array}{r}
 x^2 + x - 12 \\
 x^2 + i \quad \overline{) \quad x^4 + x^3 - 11x^2 + x - 12} \\
 \underline{x^4 \quad \quad \quad + x^2} \\
 x^3 - 12x^2 + x - 12 \\
 \underline{x^3 \quad \quad \quad + x} \\
 -12x^2 - 12 \\
 \underline{-12x^2 - 12} \\
 0
 \end{array}$$

(2)

1 division.
1 factor.

$$\therefore P(x) = (x+i)(x-i)(x+4)(x-3)$$

$$\begin{aligned}
 (f) \quad (a+ib)^2 &= 16-30i \\
 a^2 - b^2 + 2abi &= 16-30i \\
 a^2 - b^2 &= 16 \quad 2ab = -30 \\
 b &= -\frac{15}{a} \\
 \therefore a^2 - \frac{225}{a^2} &= 16 \\
 \therefore a^4 - 16a^2 - 225 &= 0 \\
 \therefore a^2 &= \frac{16 \pm \sqrt{256 - 4 \times 1 \times -225}}{2} \\
 &= \frac{16 \pm 34}{2} \\
 &= 25 \text{ or } -9 \\
 \therefore a &= \pm 5 \quad b = \mp 3
 \end{aligned}$$

(2)

1 for eqn.

1 for solution

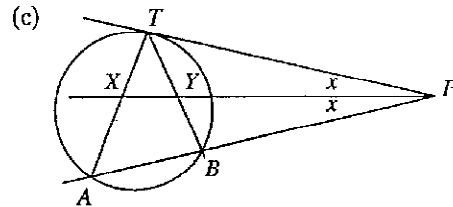
$\therefore$ Square roots are $5-3i$ and $-5+3i$

$$\begin{aligned}
\boxed{1} \quad 2. \quad (a) \quad (i) \quad \text{L.H.S} &= |z_1 + z_2|^2 + |z_1 - z_2|^2, \\
&= (z_1 + z_2)(\overline{z_1 + z_2}) + (z_1 - z_2)(\overline{z_1 - z_2}), \\
&= z_1\overline{z_1} + z_1\overline{z_2} + z_2\overline{z_1} + z_2\overline{z_2} + z_1\overline{z_1} - z_1\overline{z_2} - z_2\overline{z_1} + z_2\overline{z_2}, \\
&= 2z_1\overline{z_1} + 2z_2\overline{z_2}, \\
&= 2|z_1|^2 + 2|z_2|^2, \\
&= 2(|z_1|^2 + |z_2|^2). \quad \checkmark
\end{aligned}$$

$\boxed{2}$ (ii) The sum of the squares of the diagonals of a parallelogram is equal to the sum of the squares of the sides. $\checkmark\checkmark$

$$\begin{aligned}
\boxed{2} \quad (b) \quad (i) \quad \text{Let } z &= r \operatorname{cis} \theta, \\
\text{then } z^4 &= r^4 \operatorname{cis} 4\theta. \\
\text{i.e., } r^4 \operatorname{cis} 4\theta &= -1, \\
&= \operatorname{cis}(\pi + 2n\pi). \\
r &= 1, \quad 4\theta = \pi + 2n\pi, \\
\theta &= (2n+1)\frac{\pi}{4}, \\
&= \pm\frac{\pi}{4}, \quad \pm\frac{3\pi}{4}. \\
\therefore z &= \frac{1}{\sqrt{2}}(1 \pm i), \quad -\frac{1}{\sqrt{2}}(1 \pm i). \quad \checkmark\checkmark
\end{aligned}$$

$$\begin{aligned}
\boxed{2} \quad (ii) \quad z^4 &= \left(z + \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)\left(z + \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)\left(z - \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)\left(z - \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right), \\
&= \left(z^2 + \frac{2z}{\sqrt{2}} + \frac{1}{2} + \frac{1}{2}\right)\left(z^2 - \frac{2z}{\sqrt{2}} + \frac{1}{2} + \frac{1}{2}\right), \\
&= (z^2 + \sqrt{2}z + 1)(z^2 - \sqrt{2}z + 1). \quad \checkmark\checkmark
\end{aligned}$$



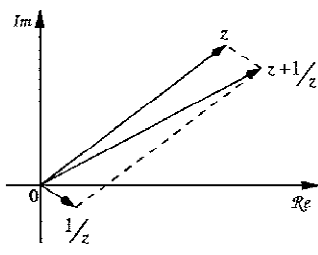
$\boxed{1}$ (i) $\angle PTB = \angle TAB$ (Angle between tangent and chord is equal to the angle in the alternate segment.) $\checkmark$

$\boxed{2}$ (ii) $\angle TPY = \angle XPA$ (PX bisects $\angle TPA$),
 $\angle PTY = \angle XAP$ (as in (i)),
 $\therefore \triangle TPY \sim \triangle XAP$ (equiangular). $\checkmark$
 $\angle TYP = \angle XAP$ (corresp. angles in similar $\triangle$'s),
 $\therefore \angle TYP = \angle XAP$ (supplementary to equal angles),
 $\therefore \triangle TXY$ is isosceles (base angles equal).
 $\therefore TX = TY$ (opposite equal angles of isosceles $\triangle$). $\checkmark$

$$\begin{aligned}
\boxed{1} \quad (iii) \quad \frac{TY}{XA} &= \frac{TP}{PA} \quad (\text{corresp. sides of similar } \triangle\text{'s}), \\
\text{but } TX &= TY, \\
\therefore \frac{TX}{XA} &= \frac{TP}{PA}. \quad \checkmark
\end{aligned}$$

- [1] (d) (i) Let $P(x) = (x - \alpha)^m \cdot Q(x)$,
 $P'(x) = m(x - \alpha)^{m-1} \cdot Q(x) + (x - \alpha)^m \cdot Q'(x)$,
 $= (x - \alpha)^{m-1} (m \cdot Q(x) + (x - \alpha) \cdot Q'(x))$,
 $= (x - \alpha)^{m-1} \cdot R(x)$. $\checkmark$
 i.e., if $P(x)$ has a root of multiplicity m , then $P'(x)$ has a root of multiplicity $m - 1$.

- [2] (ii) $P(x) = 5x^5 - 3x^3 + k$.
 $P'(x) = 25x^4 - 9x^2$,
 $= x^2(25x^2 - 9)$,
 $= x^2(5x - 3)(5x + 3)$.
 $\therefore$ The repeated positive root is $x = \frac{3}{5}$. $\checkmark$
 So $k = 3\left(\frac{3}{5}\right)^5 - 5\left(\frac{3}{5}\right)^3$,
 $= \frac{405 - 243}{625}$,
 $= \frac{162}{625}$. $\checkmark$

- [1] (e) (i) For a rhombus, $|z| = \left| \frac{1}{z} \right|$.
 $z\bar{z} = \frac{1}{z\bar{z}}$,
 $x^2 + y^2 = \frac{1}{x^2 + y^2}$,
 $x^4 + 2x^2y^2 + y^4 = 1$,
 $(x^2 + y^2)^2 = 1$,
 $x^2 + y^2 = 1, (x^2 + y^2 \neq -1)$,
 i.e., $|z| = 1$. $\checkmark$
- 

- [2] (ii) For a square, $\arg z - \arg \frac{1}{z} = \pm \frac{\pi}{2}$, $\checkmark$
 or $2(\arg z) = \frac{\pi}{2} + n\pi$,
 $\arg z = \pm \frac{\pi}{4}$ or $\pm \frac{3\pi}{4}$.
 $\therefore$ for a square, $|z| = 1$, $\arg z = \pm \frac{\pi}{4}$ or $\pm \frac{3\pi}{4}$. $\checkmark$

- [1] (f) (i) With 6 dots, we can have only 3 red, 2 white, and 1 blue.
 $\therefore$ Number of codes $= \frac{6!}{3!2!1!}$,
 $= \frac{6 \cdot 5 \cdot 4}{2 \cdot 1}$,
 $= 60$. $\checkmark$

- [2] (ii) Extra ways with 5 dots are: $\checkmark$
 $3 \text{ red} + 2 \text{ white} = \frac{5!}{3!2!}$, $3 \text{ red} + 1 \text{ white} + 1 \text{ blue} = \frac{5!}{3!1!1!}$, $2 \text{ red} + 2 \text{ white} + 1 \text{ blue} = \frac{5!}{2!2!1!}$,
 $= \frac{5 \cdot 4}{2 \cdot 1}$, $= 20$, $= \frac{5 \cdot 4 \cdot 3}{2 \cdot 1}$,
 $= 10$, $= 30$.
 $\therefore$ The number of codes with only 5 dots is 60. $\checkmark$

QUESTION 3

(a) (i) $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$ (A) (D.H.T).

Also, $(\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4 \cos^3 \theta \cdot i \sin \theta + 6 \cos^2 \theta (i \sin \theta)^2 + 4 \cos \theta (i \sin \theta)^3 + (i \sin \theta)^4$

$$= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta.$$

— (B)

Equating real parts in (A) & (B)

$$\begin{aligned} \cos 4\theta &= \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \\ &= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2 \\ &= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta \end{aligned}$$

$$\therefore [\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1] \text{ — (C)}$$

ii) Now $16z^4 - 16z^2 + 1 = 0$

$$\Rightarrow 8z^4 - 8z^2 + 1 = 0$$

In (C) let $\cos 4\theta = \frac{1}{2}$

$$4\theta = 2k\pi \pm \frac{\pi}{3}$$

$$\Rightarrow 4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$$

$$\begin{aligned} \text{Hence } z &= \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}, \cos \frac{11\pi}{12} \\ &= \pm \cos \frac{\pi}{12}, \pm \cos \frac{5\pi}{12} \end{aligned}$$

Now $S_2 = -\cos^2 \frac{\pi}{12} - \cos^2 \frac{5\pi}{12} = -1$

$$\therefore S_2 = \cos^2 \frac{\pi}{12} + \cos^2 \frac{5\pi}{12} = 1$$

and $S_4 = \cos^2 \frac{\pi}{12} \cdot \cos^2 \frac{5\pi}{12} = \frac{1}{16}$

Form a quadratic

$$A^2 - S_2 A + S_4 = 0$$

$$A^2 - A + \frac{1}{16} = 0$$

$$\begin{aligned} A &= \frac{1 \pm \sqrt{1 - \frac{1}{4}}}{2} \\ &= \frac{2 \pm \sqrt{3}}{4} \end{aligned}$$

$$\therefore \left| \cos \frac{\pi}{12} = \frac{1}{2} \sqrt{2+\sqrt{3}} \right| + \left| \cos \frac{\pi}{12} = \frac{1}{2} \sqrt{2-\sqrt{3}} \right|$$

NB ($\cos \frac{\pi}{12} > \cos \frac{\pi}{14} > 0$).

(b) given $x^3 - 2x + 3 = 0$ (A)

Let $X = \frac{1}{x}$ or $x = \frac{1}{X}$ in (A)

$$\left(\frac{1}{x}\right)^3 - 2\left(\frac{1}{x}\right) + 3 = 0$$

$$1 - 2x^2 + 3x^3 = 0 \quad \text{or} \quad 3x^3 - 2x^2 + 1 = 0 \quad (13)$$

Hence $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = -\frac{b}{a}$ in (13)

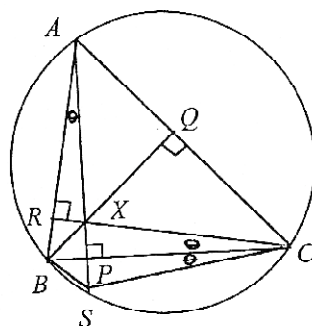
$$= -\frac{-2}{3}$$

$$= \frac{2}{3}$$

OR $x^3 - \frac{2}{3}x^2 + \frac{1}{3} = 0$

both
sides
multiplied

(c) (i)



(ii) Aim: To prove that $XP = PS$.

Construction Join BS and CS.

Proof $\angle ARC = \angle APC = 90^\circ$ (data).

$\therefore ARPC$ is a cyclic quadrilateral
($\angle$ s equal angles or same
side of interval $\Rightarrow$ pts are
concyclic).

$\therefore \angle RAP = \angle RCP = \theta$ (angles in the
same segment
standing on same
arc are equal)

also $\angle BAS = \angle BCS = \theta$ (" " ")

$\therefore$ In $\triangle XCS$

$\angle XCP = \angle SCP = \theta$ (proved)

$\angle XPC = \angle SPC = 90^\circ$ (data)

PC is common

$\therefore \triangle XPC \equiv \triangle SPC$ (AAS) $\therefore \underline{XP = PS}$ (conclusion
sides of
triangles
are equal)

(d) (i) $x^3 + ax + b = 0$ where $a, b \in \mathbb{R}, a > 0$.

(3)

Let $P(x) = x^3 + ax + b$ — (A)

$P'(x) = 3x^2 + a$ — (B)

Clearly $P'(x) > 0$ for all x .

$\therefore$ no stationary pts $\Rightarrow$ 1 Real Root.

(ii) For 2 equal roots.

$P'(x) = P(x) = 0$ for some x .

now in (B) if $3x^2 + a = 0$
 $x^2 = -\frac{a}{3}$ NB. ($a < 0$).

$\therefore x = \pm \sqrt{-\frac{a}{3}}$

Sub in (A) $-\frac{a}{3} \times \pm \sqrt{-\frac{a}{3}} + a \times \pm \sqrt{-\frac{a}{3}} + b = 0$

$\therefore \pm \frac{2}{3} a \sqrt{-\frac{a}{3}} = -b$

squaring $-\frac{4}{27} a^3 = b^2$

$\therefore \boxed{4a^3 + 27b^2 = 0}$

(c) Let the statement be

$$S(n) = 1 + 2q + 3q^2 + \dots + nq^{n-1} = \frac{1 - (n+1)q^n + nq^{n+1}}{(1-q)^2}$$

STEP I. Assume $S(1)$ is true.

$$\text{LHS} = 1 \quad \text{RHS} = \frac{1 - 2q + q^2}{(1-q)^2} = 1 \quad \therefore S(1) \text{ is true.}$$

STEP II. Assume $S(k)$ is true.

$$\text{i.e. } 1 + 2q + 3q^2 + \dots + kq^{k-1} = \frac{1 - (k+1)q^k + kq^{k+1}}{(1-q)^2}$$

STEP III R.T.P. $S(k+1)$ is true

$$\text{i.e. } 1 + 2q + 3q^2 + \dots + (k+1)q^k = \frac{1 - (k+2)q^{k+1} + (k+1)q^{k+2}}{(1-q)^2}$$

$$\text{LHS} = \frac{1 - (k+1)q^k + kq^{k+1} + (k+1)q^k}{(1-q)^2}$$

$$= \frac{1 - (k+1)q^k + kq^{k+1} + (k+1)(1-q)^2 q^k}{(1-q)^2}$$

$$= \frac{1 - (k+1)q^k + kq^{k+1} + (1-2q+q^2)(k+1)q^k}{(1-q)^2}$$

$$= \frac{1 - (k+1)q^k + kq^{k+1} + (k+1)q^k - 2(k+1)q^{k+1} + (k+1)q^{k+2}}{(1-q)^2}$$

$$= \frac{1 + q^{k+1}(k-2k-2) + (k+1)q^{k+2}}{(1-q)^2}$$

$$= \frac{1 - (k+2)q^{k+1} + (k+1)q^{k+2}}{(1-q)^2}$$

$$= \text{RHS.}$$

CONCLUSION

We have shown that $S(n)$ is true for $n=1$. $\therefore$ by the Principle of induction $S(n)$ is true for all positive integers.